

CHEAT SHEET

$$\frac{P}{1-n \cdot d}$$

$$\sum_{i=1}^n (1+r)^i$$

$$C = P \frac{Q}{1+r}$$

$$EOQ = \sqrt{\frac{2 \times \text{Annual Demand}}{\text{Ordering Cost}}}$$

PROBABILITY AND STATISTICS



EVENTS

Single EVENT

$$P(E) = \frac{n(E)}{n(S)} = \frac{\text{NUMBER of fav}}{\text{Total outcomes}}$$

$$1) 0 \leq P(A) < 1$$

$$2) P(\text{sure event}) = 1$$

$$3) P(\text{Impossible event}) = 0$$

$P(E^c) = \text{complementary event}$

$$P(E) + P(E^c) = 1$$

$$P(E^c) = 1 - P(E)$$

Compound EVENTS

"Something common"

NON EXCLUSIVE



"Nothing common"

EXCLUSIVE



	E_2	E_2	$\bar{E}_2$
E_1	$E_1 \cap E_2$ (both occur)	$\bar{E}_1 \cap E_2$ (only E_2)	
$\bar{E}_1$	$E_2 \cap \bar{E}_1$ (only E_2)	$\bar{E}_1 \cap \bar{E}_2$ (Neither E_1 Nor E_2)	

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) \text{ where } E_1, E_2 \text{ are disjoint}$$

WHERE $P(E_1 \cap E_2) = P(\text{both occur}) = P(\text{simultaneously occur})$

$$\# 1) \quad P(E_1 \cap E_2) = \frac{n(E_1 \cap E_2)}{n(S)}$$

$$\# 2) \quad P(E_1 \cap \bar{E}_2) = P(\text{only } E_1 \text{ occur}) = P(E_1) - P(E_1 \cap E_2)$$

$$\# 3) \quad P(\bar{E}_1 \cap E_2) = P(\text{only } E_2 \text{ occur}) = P(E_2) - P(E_1 \cap E_2)$$

4) If E_1 and E_2 ARE independent events

$$P(E_1 \cap E_2) = P(E_1) P(E_2)$$

$$\text{common} = P(E_1) P(E_2)$$

prob.

5) If E_1, E_2 ARE independent events $\bar{E}_1, \bar{E}_2$ ARE also independent

$$P(\bar{E}_1 \cap \bar{E}_2) = P(\bar{E}_1) P(\bar{E}_2)$$

For THREE EVENTS:- E_1, E_2, E_3 ARE independent events

$$P(E_1 \cap E_2 \cap E_3) = P(E_1) P(E_2) P(E_3)$$



$$E_1 \cap E_2 \cap E_3$$

$$E_1 \cap E_2 \cap \bar{E}_3$$

$$E_1 \cap \bar{E}_2 \cap E_3$$

$$E_1 \cap \bar{E}_2 \cap \bar{E}_3$$

$$\bar{E}_1 \cap E_2 \cap E_3$$

$$E_1 \cap E_2 \cap \bar{E}_3$$

$$\bar{E}_1 \cap E_2 \cap \bar{E}_3$$

$$E_1 \cap \bar{E}_2 \cap \bar{E}_3$$



(something is common)

$$P(E_1 \cup E_2 \cup E_3) = P(\text{at least one occurs})$$

$$\Rightarrow \frac{P(E_1) + P(E_2) + P(E_3) - P(E_1 \cap E_2) - P(E_2 \cap E_3) - P(E_3 \cap E_1) + P(E_1 \cap E_2 \cap E_3)}{}$$

$P(\text{at least two events occur})$

$$= P(E_1 \cap E_2) + P(E_2 \cap E_3) + P(E_3 \cap E_1) - 2P(E_1 \cap E_2 \cap E_3)$$

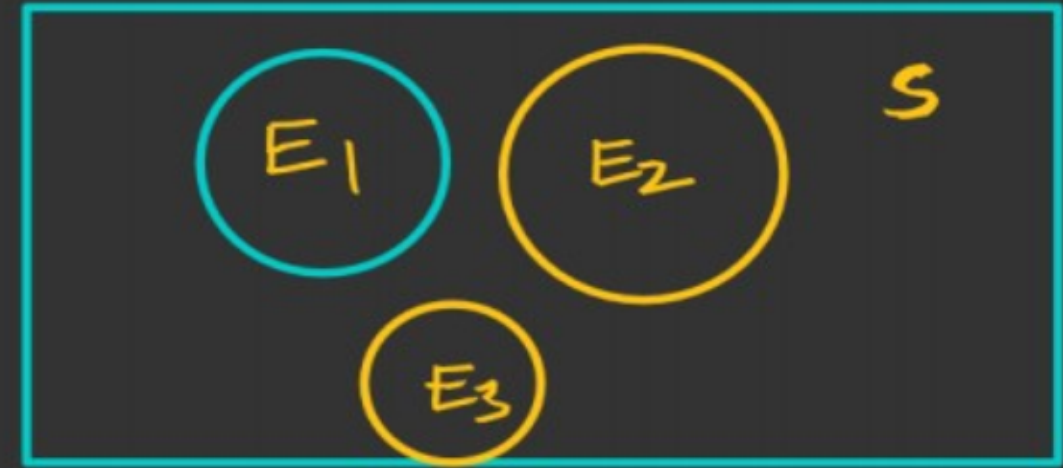
● $P(\text{exactly two events})$

$$= P(E_1 \cap E_2) + P(E_2 \cap E_3) + P(E_3 \cap E_1) - 3P(E_1 \cap E_2 \cap E_3)$$



● E_1, E_2, E_3 ARE EXCLUSIVE EVENTS (Nothing is common)

$$P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3)$$



● $P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2) = S_1 - S_2$

$$P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3) - P(E_1 \cap E_2) - P(E_2 \cap E_3) - P(E_3 \cap E_1) + P(E_1 \cap E_2 \cap E_3)$$

$$\vdots = S_1 - S_2 + S_3$$

$$P(E_1 \cup E_2 \cup E_3 \cup E_4) = S_1 - S_2 + S_3 - S_4$$

$$P(E_1 \cup E_2 \cup E_3 \dots E_n) = S_1 - S_2 + S_3 - S_4 \dots (-1)^{n-1} S_n$$

(POINCARÉ INEQUALITY)

$$P(E_1 \cup E_2 \cup E_3 \dots E_n)$$

(Hat Matching Problem)

conditional Probability :- $P(A \text{ given } B) = \frac{P(A \cap B)}{P(B)} = \frac{\text{common}}{P(B)}$

$$P(B \text{ given } A) = \frac{P(A \cap B)}{P(A)} \quad \text{WHERE}$$

$P(A) \neq 0$
 $P(B) \neq 0$

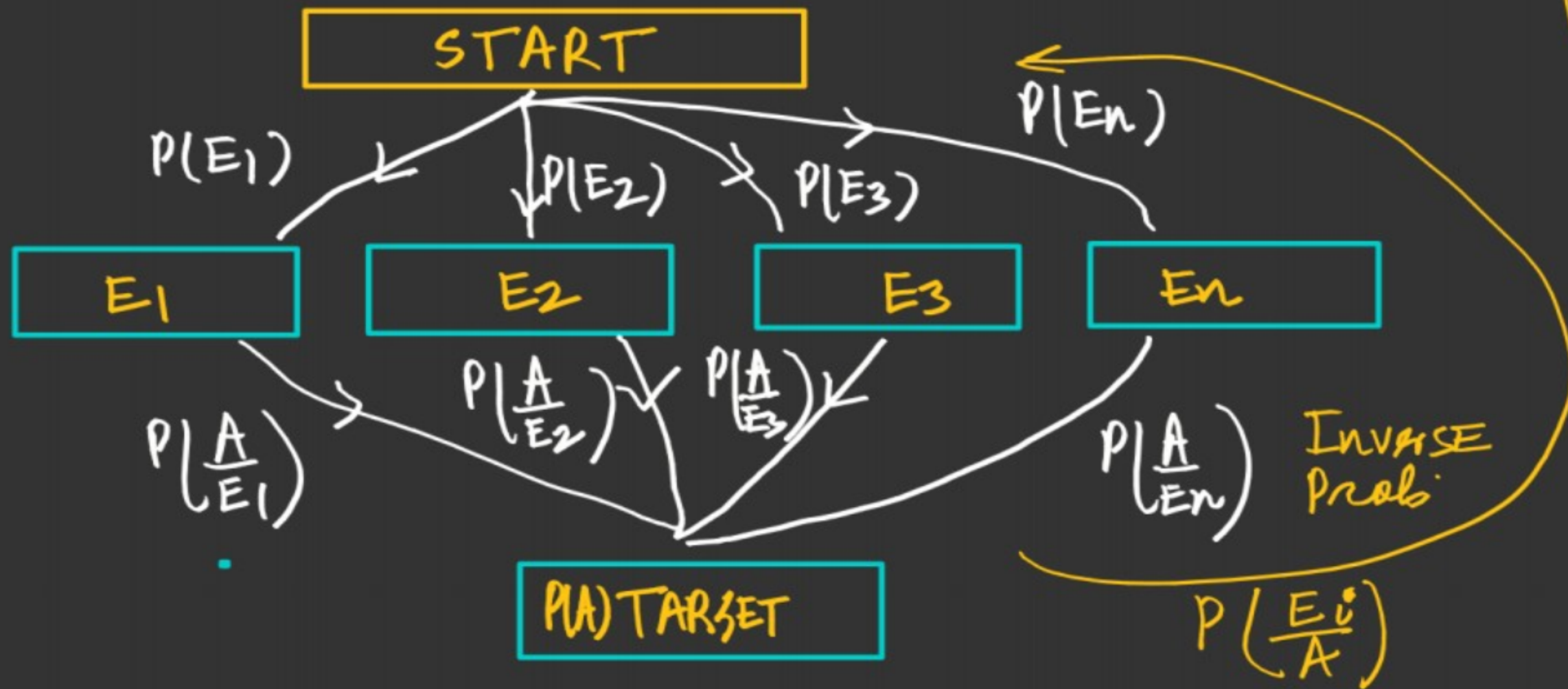
If A and B ARE Dependent EVENTS

$$P(A \cap B) = P(A)P\left(\frac{B}{A}\right)$$

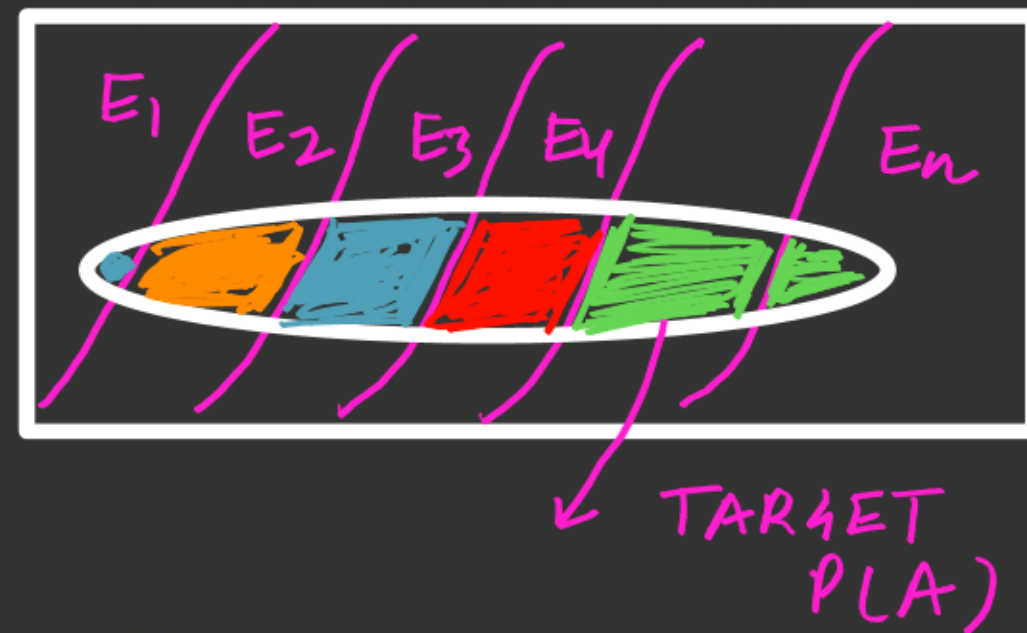
X given Y means

$\xrightarrow{X}$ — Future
Y — PAST

Total probability / Bayes Theorem :-



$$P\left(\frac{E_i}{A}\right) = \frac{P(E_i) P\left(\frac{A}{E_i}\right)}{\sum_{l=1}^n P(E_l) P\left(\frac{A}{E_l}\right)}$$



Total prob.

$$P(A) = P(E_1)P\left(\frac{A}{E_1}\right) + P(E_2)P\left(\frac{A}{E_2}\right) + \dots + P\left(\frac{A}{E_n}\right)P(E_n)$$

$$= \sum_{l=1}^n P(E_l)P\left(\frac{A}{E_l}\right)$$

Random Variable

Integer
(Arrival) ✓



DISCRETE random Vari

prob. MASS function $P[X=x]$ (Integer)

cumulative Distribution Function

$$F(x) = \sum_{\text{START}}^x P(X \leq x)$$

valid prob. MASS function

$$\sum_{i=1}^n P(X=x_i) = 1$$

$$\# E(X) = \sum_{i=1}^n x_i P(X=x_i)$$

$$\# V(X) = E(X^2) - [E(X)]^2$$



Interval
(waiting time) ✓

continuous Random Variable

prob. DENSITY function $f(x)$

valid prob. DENSITY function $\int_{-\infty}^{\infty} f(x) dx = 1$

cumulative Distribution function

$$cdf = \int_{\text{START}}^x f(x) dx$$

expected value $E(X) = \int_a^b x f(x) dx$

Variance $V(X) = E(X^2) - [E(X)]^2$ where $E(X^2) = \int_a^b x^2 f(x) dx$

Standard deviation = $\sqrt{\text{variance}}$

variance can't be Negative

Properties of Expected value E / Variance V / Covariance C / Correlation

A) X and Y ARE independent R.V

$$E(X+Y) = E(X) + E(Y)$$

$$B) E(cX) = cE(X)$$

$$C) E(aX+bY) = aE(X) + bE(Y)$$

$$D) V(-X) = (-1)^2 V(X)$$

$$E) \text{cov}(X, Y) = E(XY) - E(X)E(Y)$$

$$F) \text{cov}(X, X) = V(X)$$

$$G) \text{cov}(Y, Y) = V(Y)$$

$$H) \text{cov}(X+Y, X+Y) = V(X) + V(Y) + 2\text{cov}(X, Y)$$

$$(I) \text{cov}(X, Y) = 0$$

X and Y ARE independent

$$(J) \text{cov}(aX+b) = a\text{cov}(X)$$

$$(K) \text{cov}(aY+b) = a\text{cov}(Y)$$

$$(L) \text{cov}(aX+b, cX+d) = ac\text{cov}(X, Y)$$

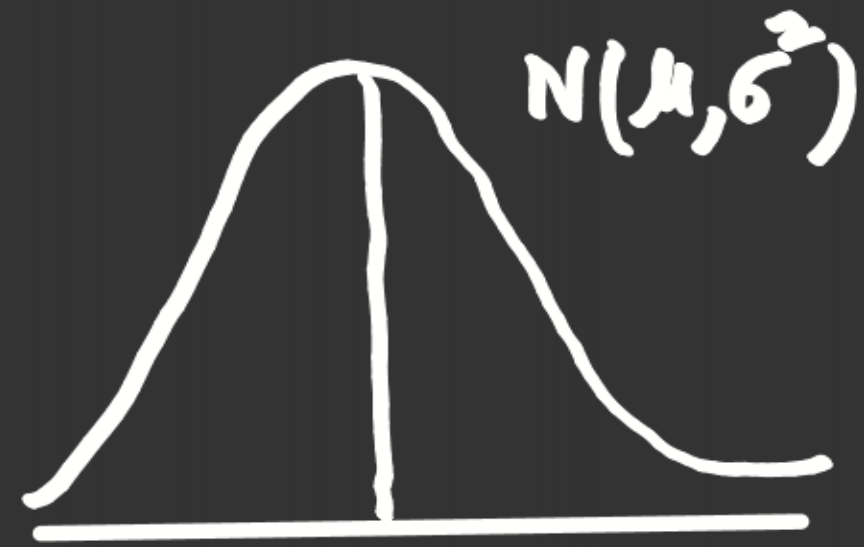
$$(M) \text{cov}(aX+bY) = a^2 V(X) + b^2 V(Y) + 2ab\text{cov}(X, Y)$$

Type	Probab. Function	MEAN	Variance	S.D	Type	
• Bernoulli (p) ①	$P(X=x) = p^x (1-p)^{1-x}$ $x=0,1$	p	pq	$\sqrt{pq}$	DISCRETE	only ONE (SUCCESS)
• Binomial (n, p) ②	$P(X=r) = {}^n C_r p^r q^{n-r}$ $r=0,1,2, \dots$	np	npq	$\sqrt{npq}$	DISCRETE	(ONE OR MORE SUCCESS)
Geometric (p) ①	$P(X=x) = q^{x-1} \cdot p$ $x=1,2, \dots$	$\frac{1}{p}$	$\frac{q}{p^2}$	$\sqrt{\frac{q}{p^2}}$	DISCRETE	final success (only)
Negative Binomial ②	$P(X=r\text{th success})$ $= {}^{n+r-1} C_{r-1} p^r (1-p)^n$	$\frac{rq}{p}$	$\frac{rq}{p^2}$	$\sqrt{\frac{rq}{p^2}}$	DISCRETE	ONE OR MORE + final success
Poisson (λ) ①	$P(X=r) = \frac{e^{-\lambda} \lambda^r}{r!}$ $r=0,1,2, \dots$	λ	λ	$\sqrt{\lambda}$	DISCRETE	AVERAGE No. of success
Uniform (a, b) ②	$P(X=x) = \frac{1}{(b-a+1)}$	$\frac{a+b}{2}$	$\frac{(b-a+1)^2}{12}$	$\sqrt{\frac{(b-a+1)^2}{12}}$	DISCRETE	Prob. of event in every part (SAME)

Conti. Prob. distribution	Function	MEan	Variance	Mode	Median
# Exponential ① distribution (ONE SUCCESS - given waiting time)	$P(X=x) = \lambda e^{-\lambda x} \quad x > 0, \lambda > 0$ cumulative distribution function $P(X \leq x) = \int_0^x \lambda e^{-\lambda x} = 1 - e^{-\lambda x}$	$\mu = \frac{1}{\lambda}$	$V(X) = \frac{1}{\lambda^2}$	Mode = 0	Median = $\frac{\ln 2}{\lambda}$
# Gaussian distribution ② (N > 30)	$N(\mu, \sigma^2) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(X-\mu)^2}{2\sigma^2}}$ $-\infty \leq \mu \leq \infty \quad \sigma > 0$	μ	σ^2	μ	μ
# Gamma distribution ② (ONE or more SUCCESS in given waiting time)	$f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} \quad x > 0$ $\alpha = \text{NO. of SUCCESS}$ $\lambda = \text{PARAMETER}$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$	$\frac{\alpha-1}{\lambda}$ $\alpha > 1$	X
# Chi square distribution ② $\alpha = \frac{\nu}{2} \quad \lambda = \frac{1}{2}$ cumulative prob. (Gamma distribution)	$f(x) = \frac{\left(\frac{1}{2}\right)^{\nu/2} x^{\nu/2-1} e^{-\frac{x}{2}}}{\Gamma\left(\frac{\nu}{2}\right)}$ Degree of freedom = $(n-1)$	$E(X) = \nu$ MEan $= \nu$ $= \text{degree of freedom}$	$V(X) = 2\nu$	X	X

Standard Gaussian Distribution :-

$$N(\mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad -\infty \leq \mu \leq \infty \quad \sigma \geq 0$$



$$\mu = 0 \quad \sigma = 1 \quad z = \frac{x - \mu}{\sigma}$$

$$N(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

z = statistical parameter

- # Symmetric about origin
- # bell shaped
- # NEVER touches or crosses horizontal axis



$$P(-1 \leq z \leq 1) = 0.6834$$

$$P(-2 \leq z \leq 2) = 0.9545$$

$$P(-3 \leq z \leq 3) = 0.9971$$

Standard
 $N(z)$ = gaussian

Bivariate Random Variables

DISCRETE random Variable

valid prob. MASS function = 1

$$\sum \sum P[X=x_i \wedge Y=y_j] = 1$$

cumulative distribution Function

$$F(x, y) = P[X \leq x, Y \leq y]$$

Marginal probability function

$$P(X=x) = \sum_{j=1}^n P[X=x_i \wedge Y=y_j]$$

$$P(Y=y) = \sum_{i=1}^n P[X=x_i \wedge Y=y_j]$$

continuous Random Variable

joint prob. Density function

$$\int_{x=a}^b \int_{y=c}^d f(x, y) dy dx = 1$$

$$F(x, y) = \int_{\text{START}}^x \int_{\text{START}}^y f(x, y) dy dx$$

$$f(x) = \int f(x, y) dy$$

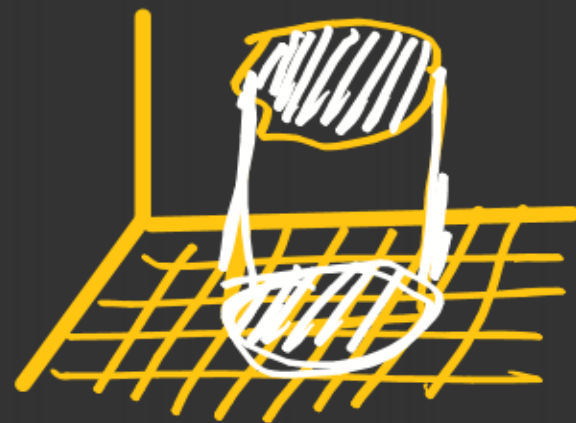
$$f(y) = \int f(x, y) dx$$

USE vertical strip

$$y_1 = g(x) \quad y_2 = h(x)$$

USE Horizontal strip

$$x_1 = f(y) \quad x_2 = g(y)$$



Prob = volume

conditional prob. distribution

$$P(Y \text{ given } X) = P(Y=y/X=x)$$

$$P\left(\frac{Y}{X}\right) = \frac{P(Y=y \cap X=x)}{P(X=x)} = \frac{\text{joint prob.}}{\text{marginal prob. of } X}$$

$$P\left(\frac{X}{Y}\right) = \frac{P(X=x \cap Y=y)}{P(Y=y)} = \frac{\text{joint prob.}}{\text{marginal prob. of } Y}$$

If X and Y ARE independent

$$P(X=x \cap Y=y) = P(X=x) P(Y=y)$$

$$\# E(X) = \sum x P(X=x)$$

$$E(Y) = \sum y P(Y=y)$$

$$E(XY) = \sum \sum xy P[X=x \cap Y=y]$$

$$\# f\left(\frac{Y}{X}\right) = \frac{f(x,y)}{f(x)} = \frac{\text{joint prob.}}{\text{marginal of } X}$$

$$f\left(\frac{X}{Y}\right) = \frac{f(x,y)}{f(y)} = \frac{\text{joint prob.}}{\text{marginal of } Y}$$

If X and Y ARE independent

$$f(x,y) = f(x) f(y)$$

$$\# E(X) = \int_a^b x f(x) dx$$

$$E(Y) = \int_a^b y f(y) dy$$

$$E(XY) = \int_a^b \int_c^d xy f(x,y) dy dx$$

conditional expectation

$$E(X \text{ GIVEN } Y) = E\left(\frac{X}{Y}\right)$$

$$E(Y \text{ GIVEN } X) = E\left(\frac{Y}{X}\right)$$

$$E\left(\frac{X}{Y}\right) = \sum_{i=1}^n x_i \cdot \frac{P[X=x_i \wedge Y=y]}{P[Y=y]}$$

$$E\left(\frac{Y}{X}\right) = \sum_{j=1}^n y_j \cdot \frac{P[X=x \wedge Y=y_j]}{P[X=x]}$$

conditional expectation

$$E\left(\frac{X}{Y}\right) = \int x \frac{f(x,y)}{f(y)} dx \quad (\text{Horizontal strip})$$

$$E\left(\frac{Y}{X}\right) = \int y \frac{f(x,y)}{f(x)} dy \quad (\text{Vertical strip})$$

$$\neq E\left(E\left(\frac{Y}{X}\right)\right) = E(Y)$$

$$E\left(E\left(\frac{X}{Y}\right)\right) = E(X)$$

$$V(X) = V\left(E\left(\frac{X}{Y}\right)\right) + E\left(V\left(\frac{X}{Y}\right)\right)$$

$$V(Y) = V\left(E\left(\frac{Y}{X}\right)\right) + E\left(V\left(\frac{Y}{X}\right)\right)$$